IN0013 LAB 1

In this week we have discussed the origin of Computer. The discussion spans from the initial need for computation during the ancient time to the incorporation of electronic in our day-to-day computers. Modern computers are getting smaller and faster that we can access it at any time and anyhow we want.

Task 1: Mind Map

Create a mind map for the origin of computers all the way to the current state. This mind map will eventually help you review this topic for quiz and labs.

Abacus (2700-3000 BC)

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Astrolabe, Antikythera, Al-Jazari's Castle Clock (3000 BC – 1000 AD)

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Computer Job Title (1613)

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Leibniz's step reckoner (1624)

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Babbage's Analytical engine (1822)

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Census Problem (1880)

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Hollerith's machine and Tabulating Machine Company (1924)

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Colussus MK1 (1943)

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Harvard Mark 1 (1944)

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ENIAC (1946)

Task 2: Binary system

Who invented the binary system? Talk a bit more regarding this person? What else is this person famous for?

Leibniz invented it. He was a German mathematician and philosopher and invented the Stepped Reckoner which was a mechanical calculator. It was able to do basic arithmetic operations.

Task 3: Vacuum vs transistor tube

Explain in your own key words what the vacuum and transistor tubes are. Which one of them is better? Why?

Vacuum tubes are electronic components that control electricity using heated filaments. It produces heat, large, fragile and consumed a lot of power.

Transistor tubes are much smaller and are solid state components. It is better because it is more durable, smaller in size and also more energy efficient.

Task 4: ENIAC

Why was the ENIAC an important machine? How does the input and output work for ENIAC? Does it even have a monitor to show output?

ENIAC was the first general-purpose electronic programmable computer. It performs thousands of calculations per second. The input was punch cards and switches where it required manual rewiring. The output printed results from punch cards. It did not have a monitor and the output was paper and lights.

Task 5: Explain how Antykhitera and aljazari's castle clock works? How do they mechanically work?

It used interlocking gears and predicted solar and lunar eclipses, planetary movements and calendar cycles.

It was a water powered clock which used water flow and gears. It was the first programmable automated machine.

Extra Task (optional):

"Moore's Law states that the number of transistors on a microchip doubles every two years. The law claims that we can expect the speed and capability of our computers to increase every two years because of this, yet we will pay less for them. Another tenet of Moore's Law asserts that this growth is exponential. The law is attributed to Gordon Moore, the co-founder and former CEO of Intel."

will this still true let's say in 2030?

Probably not true. There are physical limits of a transitor size and heat and power consumptions.